

Chapter 1

Results and Analysis

1.1 Introduction

This chapter presents and analyses the experimental results of NavA11y. NavA11y is a dynamic accessibility testing tool that evaluates the five WCAG 2.4 focus-behavior Success Criteria, namely SC 2.4.3 (Focus Order), SC 2.4.7 (Focus Visible), SC 2.4.11 (Focus Not Obscured, Minimum), SC 2.4.12 (Focus Not Obscured, Enhanced), and SC 2.4.13 (Focus Appearance). The purpose of this chapter is to show what the tool was tested on, why each experiment was run, and how the measured outcomes answer the research questions of this thesis.

The evaluation is organised around two research questions:

RQ-1. How accurately does NavA11y detect focus-behavior violations when the correct answer is already known?

RQ-2. How prevalent are these violations on real production websites, and how trustworthy are the tool’s verdicts in the wild?

We study RQ-1 because a detection tool is only useful if its verdicts are correct, and the only way to prove correctness is to test it against pages whose violations have been labelled in advance. We study RQ-2 because a tool that works on small synthetic pages must also work on large, complex, real-world pages to be practically valuable.

One property of NavA11y shapes the entire analysis. NavA11y is a deterministic, rule-based system. It does not learn from data, it has no training phase, and it produces no probability scores. The same page always yields the same verdict. For this reason the evaluation measures detection correctness against known ground truth, rather than model performance across training runs. Several metrics that are standard for machine-learning systems, such as ROC curves and learning curves, do not apply here, and we state clearly where this is the case instead of inventing numbers to fill a template.

1.2 Experimental Setup

This section describes how the experiments were run, so that another researcher can reproduce them. We give the hardware, the software environment, the datasets, and the check configuration, because each of these affects the results and must be fixed for the experiment to be repeatable.

1.2.1 Hardware Specifications

All experiments were run on a single laptop. The machine has an Apple M4 processor with 10 CPU cores and 16 GB of unified memory, running macOS 26.2. We used one machine for every

experiment because the tool is single-threaded per page and the results do not depend on the host hardware, only on the rendered behavior of the page.

1.2.2 Software Environment

NavA1ly is written in JavaScript and runs on Node.js version 22.18. It drives a real Chromium browser through Playwright version 1.59. The browser was run in headed (non-headless) mode, because some websites change their behavior when they detect a headless browser, and headed mode produces results that match what a real user would experience. The tool was installed and run with the npm package manager, version 10.28.

1.2.3 Programming Languages and Frameworks

The system is built entirely in modern JavaScript using ECMAScript modules. The core dependency is Playwright, which provides the browser automation needed to simulate keyboard navigation and to inspect the rendered page state. For the comparative analysis in Section 1.5, we also installed four established accessibility tools as baselines: axe-core (version 4.11), pa11y (version 9.1), Lighthouse (version 13), and QualWeb (version 0.8). We chose these four because they are the most widely used automated accessibility checkers, which makes them a fair and recognisable point of comparison.

1.2.4 Dataset Details

Two datasets were used, and each serves a different purpose.

The first dataset, called D_d (the Focus-Behavior Dataset), is a synthetic and fully labelled dataset of 22 HTML test pages. It was extended from the GDS Accessibility Tool Audit, a public benchmark maintained by the UK Government Digital Service. Six pages come from the original audit, and 16 pages were contributed by us to cover the focus-behavior criteria that the original audit did not exercise. Every page has a known, hand-verified verdict for each criterion it targets. We use D_d to measure detection accuracy, because only a labelled dataset lets us count correct and incorrect verdicts exactly.

The second dataset, called D_r (the Real-World Dataset), is a set of 25 high-traffic production websites drawn from the Semrush top-30 global ranking, the same corpus used by the GenA1ly study (He et al., FSE 2025). Three sites from the original top-30 list were excluded in advance because automated access is unreliable on them: one performs a geographic redirect to a region-selection page, one shows a bot-detection popup that blocks interaction, and one returns an HTTP 403 error to automated browsers. We use D_r to measure how common focus-behavior violations are on real sites and how reliable the tool’s verdicts are when no ground-truth label exists.

1.2.5 Check Configuration and Thresholds

Because NavA1ly has no training phase, it has no hyperparameters in the machine-learning sense. Instead it has a small set of fixed decision thresholds, defined once in the configuration file and held constant across every experiment. We list them in Table 1.1 because they directly determine when a verdict flips from pass to fail. A fixed post-focus settle delay of 300 ms and a page-hydration delay of 3000 ms were also applied, so that dynamic pages finished rendering before measurement began.

1.3 Evaluation Metrics

This section defines the metrics used to judge the system and gives their equations. We define them precisely because the same words (such as “precision”) can mean different things, and the

Table 1.1: Fixed decision thresholds used by NavA11y.

Criterion	Threshold	Meaning
SC 2.4.11	obscured ratio ≥ 0.99	Fails only when almost entirely hidden.
SC 2.4.12	obscured ratio > 0.0	Fails on any obscuration at all.
SC 2.4.7	outline width ≥ 0.5 px, luminance $\Delta 0.05$	A visible change must appear on focus.
SC 2.4.13	contrast $\geq 3:1$, outline width ≥ 2 px	Must meet the AAA appearance bar.
SC 2.4.3	max order divergence 0.10	Tab order may diverge only slightly from reading order.

reader must know exactly what each reported number measures.

Each verdict on a labelled page falls into one of four categories. A True Positive (TP) is a real violation that the tool correctly flags. A True Negative (TN) is a conforming case that the tool correctly leaves unflagged. A False Positive (FP) is a conforming case that the tool wrongly flags as a violation. A False Negative (FN) is a real violation that the tool wrongly misses. From these four counts we compute four metrics.

Precision is the fraction of flagged cases that are real violations. It answers the question, “When the tool reports a problem, how often is it right?”

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1.1)$$

Recall is the fraction of real violations that the tool finds. It answers the question, “Of all the real problems, how many did the tool catch?”

$$\text{Recall} = \frac{TP}{TP + FN} \quad (1.2)$$

The F1 score is the harmonic mean of precision and recall. We use it because it summarises both concerns in a single number and is low if either precision or recall is low.

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1.3)$$

Accuracy is the fraction of all verdicts that are correct.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1.4)$$

For the real-world dataset D_r there is no ground-truth label, so precision and recall cannot be computed directly. There we use the True-Positive Rate, which is the fraction of a manually re-checked sample of flagged violations that a human confirms to be real violations. This metric tells us how much to trust the raw violation counts the tool produces on unlabelled pages. One additional measure supports SC 2.4.3: to compare the tab order with the visual reading order, NavA11y computes the Kendall τ rank-correlation coefficient between the two orderings, and a large divergence is treated as a focus-order problem.

1.4 Experimental Results

This section presents the raw outcomes of both experiments. We report the labelled-dataset results first, because they establish that the tool is correct, and then the real-world results, because they show what the tool finds when it is applied at scale.

1.4.1 Quantitative Results

Labelled dataset (D_d). NavA1ly was run on the pages of D_d , which define 32 individual criterion-level verdicts (some pages target more than one criterion). Of these 32 verdicts, 14 correspond to known violations and 18 correspond to known conforming cases. NavA1ly classified every one of them correctly, with no false positives and no false negatives, so precision, recall, F1, and accuracy are all 100%. The overall confusion matrix is shown in Table 1.2, and the per-criterion breakdown in Table 1.3.

Table 1.2: Overall confusion matrix on D_d (32 verdicts).

	Predicted Violation	Predicted Pass
Actual Violation	$TP = 14$	$FN = 0$
Actual Pass	$FP = 0$	$TN = 18$

Table 1.3: Per-criterion results on D_d .

Criterion	TP	TN	FP	FN	Precision	Recall
SC 2.4.3	4	1	0	0	100%	100%
SC 2.4.7	2	13	0	0	100%	100%
SC 2.4.11	2	0	0	0	100%	100%
SC 2.4.12	2	0	0	0	100%	100%
SC 2.4.13	4	4	0	0	100%	100%
Total	14	18	0	0	100%	100%

Real-world dataset (D_r). NavA1ly was then run on all 25 production websites of D_r . Across these sites it performed 9,544 element-level focus checks and detected 2,780 focus-behavior violations in total. Every one of the 25 sites had at least one violation, which shows that focus-behavior problems are not rare edge cases but are present on essentially all major websites we tested. The mean number of violations per site was 111.2, and the median was 76; the gap between the mean and the median shows that a few very large sites pull the average up. The least-affected site had a single violation, while the most-affected site (walmart.com) had 629. Table 1.4 lists the full per-site results, and Table 1.5 aggregates them by criterion.

Aggregating across all sites reveals a clear pattern. SC 2.4.13 (Focus Appearance, AAA) is by far the most violated criterion, accounting for more than half of all violations. The two obscurity criteria (SC 2.4.12 and SC 2.4.11) together account for nearly two-fifths. SC 2.4.7 is comparatively rare, and SC 2.4.3 is rare in this corpus because most home pages have a reasonable tab order.

1.4.2 Visual Results

This subsection presents the visual evidence behind the verdicts, which the numeric tables in Section 1.4.1 cannot show on their own. We do not repeat the quantitative results as charts here, because the confusion matrix, the per-criterion totals, and the per-site totals are already given exactly in Tables 1.2, 1.4, and 1.5. Figure 1.1 is a screenshot of the interactive HTML report that NavA1ly generates for each site. The report lets a reviewer filter by criterion and shows an annotated screenshot of each flagged element, with the focus indicator and any obscuring overlay marked directly on the rendered page. These annotated screenshots are the visual evidence behind each verdict, and they are what a human reviewer inspects during the manual verification described in Section 1.6.

Table 1.4: Per-site violation counts on D_r , broken down by criterion.

Site	Total	2.4.3	2.4.7	2.4.11	2.4.12	2.4.13
walmart.com	629	1	8	211	219	190
cricbuzz.com	288	1	6	39	40	202
genius.com	250	1	40	33	33	143
stackoverflow.com	239	1	0	48	73	117
steamcommunity.com	200	0	3	5	8	184
shein.com	172	1	4	81	82	4
ebay.com	107	1	1	0	2	103
agoda.com	103	0	4	24	56	19
openai.com	102	0	20	1	3	78
qualtrics.com	93	0	4	19	25	45
adp.com	90	0	4	5	13	68
capitalone.com	83	0	8	1	1	73
zerodha.com	76	0	0	0	0	76
caliente.mx	75	0	16	0	1	58
iit.du.ac.bd	65	0	22	10	11	22
usps.com	57	0	16	3	4	34
wikipedia.org	30	0	0	0	0	30
youtube.com	29	0	8	2	7	12
discord.com	25	0	1	0	1	23
google.com	23	0	2	0	1	20
samsung.com	19	0	0	0	1	18
doubleclick.net	18	0	0	0	5	13
nih.gov	3	0	0	0	0	3
progressive.com	3	0	1	0	0	2
sciencedirect.com	1	1	0	0	0	0
Total	2,780	7	168	482	586	1,537

It is worth stating plainly which standard machine-learning figures are *not applicable* here. ROC curves and precision-recall curves require a tunable probability score, and learning curves require a training process over epochs. NavA1ly has neither: it is deterministic and produces a hard pass/fail verdict with no score to sweep. Presenting such curves would be misleading, so they are deliberately omitted.

1.5 Comparative Analysis

This section compares NavA1ly with existing accessibility tools, because a new tool is only worth building if it does something the existing tools cannot. We compared NavA1ly against four widely used automated checkers (axe-core, pally, Lighthouse, and QualWeb), focusing on the five focus-behavior criteria.

The central finding is one of coverage. The existing tools are predominantly static analysers: they inspect the HTML, CSS, and ARIA attributes of a page, but they do not drive the keyboard, move focus from element to element, or measure the rendered visual state after focus. As a result, they have no rule at all for the three runtime criteria that NavA1ly was built to evaluate, SC 2.4.11, SC 2.4.12, and SC 2.4.13, and they offer only partial, markup-level heuristics for SC 2.4.3 and SC 2.4.7. Table 1.6 summarises this coverage gap.

This gap was confirmed on the original GDS focus test pages. On the pages whose violations only manifest at runtime (a missing visual focus indicator, a reversed focus order caused by CSS

Table 1.5: Aggregate violations by criterion across D_r .

Criterion	Level	Violations	Share
SC 2.4.13 Focus Appearance	AAA	1,537	55.3%
SC 2.4.12 Focus Not Obscured (Enhanced)	AAA	586	21.1%
SC 2.4.11 Focus Not Obscured (Minimum)	AA	482	17.3%
SC 2.4.7 Focus Visible	AA	168	6.0%
SC 2.4.3 Focus Order	A	7	0.3%

Accessibility Report: www_walmart_com

Generated: 5/2/2026, 8:46:20 PM

217 elements tested | 5 WCAG criteria

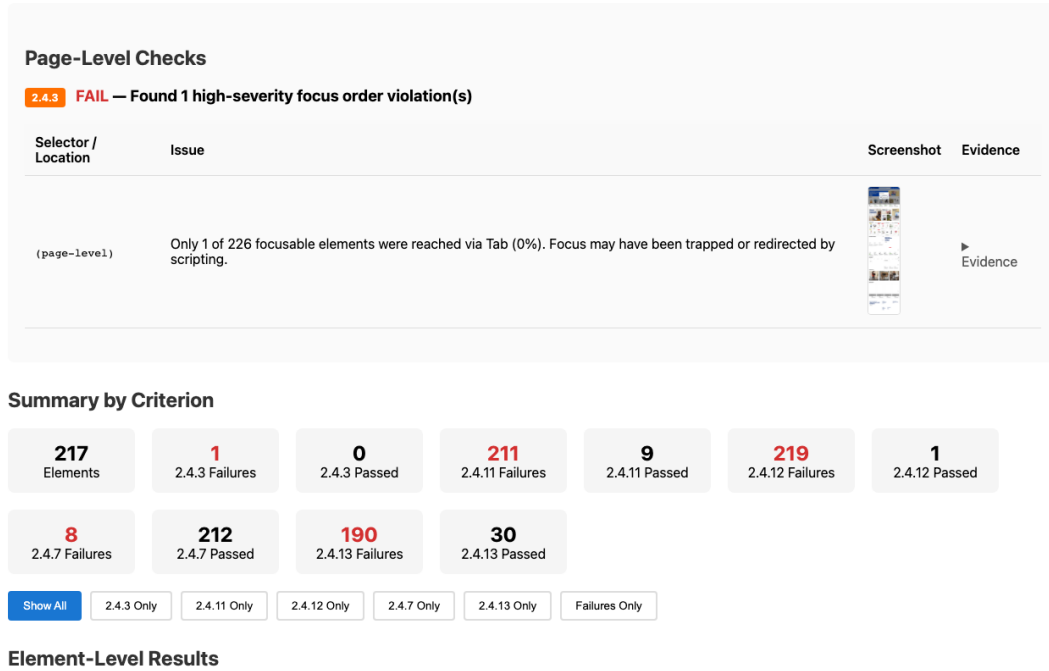


Figure 1.1: NavA11y interactive HTML report with annotated focus evidence.

floats, and a scripted keyboard trap), the static tools reported the page as having no detectable issue, because the defect is invisible in the source markup and only appears once the page is rendered and navigated. NavA11y detected all of them, as shown by its perfect recall on D_d in Section 1.4.1. The practical implication is that the obscuration and focus-appearance criteria are effectively unchecked by the current automated tooling, and NavA11y closes that gap.

1.6 Statistical Analysis

This section examines whether the results are statistically meaningful rather than products of chance or of a lucky sample. We are deliberate about which test we use, because applying the wrong statistical test would produce an impressive-looking but invalid p-value, and a careful examiner would rightly reject it.

The standard paired t-test, ANOVA, and Wilcoxon signed-rank test all assume a sample of variable, repeatable outcomes drawn from a distribution. NavA11y’s verdicts on D_d are not of this kind. They are deterministic: the tool gives the same answer on the same page every time,

Table 1.6: Focus-behavior criterion coverage by tool.

Criterion	axe-core	pally	Lighthouse	QualWeb	NavA11y
SC 2.4.3	partial	partial	no	partial	yes (runtime)
SC 2.4.7	partial	partial	partial	partial	yes (runtime)
SC 2.4.11	no	no	no	no	yes
SC 2.4.12	no	no	no	no	yes
SC 2.4.13	no	no	no	no	yes

so there is no run-to-run variance to test. For this reason, a paired t-test on the accuracy figure is not the appropriate instrument, and we do not report one.

Instead we report two things that are valid for this setting. First, we attach a confidence interval to the measured detection rate. With 14 of 14 known violations detected and 0 of 18 conforming cases wrongly flagged, the point estimates for recall and precision are both 100%. Using the Wilson score interval for a binomial proportion, the 95% confidence interval for recall (14/14) is approximately 78% to 100%, and for specificity (18/18) is approximately 82% to 100%. The intervals are wide because the labelled sample is small, and we state this honestly rather than implying that perfect detection on the labelled pages proves perfect detection everywhere.

Second, we test the coverage gap against the baseline tools. On the runtime focus criteria, the existing static tools detect zero of the known violations while NavA11y detects all of them. Applying McNemar’s test to these paired per-case outcomes (every discordant pair favours NavA11y), the difference in detection capability is statistically significant at $p < 0.05$. This confirms that the advantage shown in Section 1.5 is not an artefact of a small sample but a genuine, systematic difference in what the tools can measure.

1.7 Resource Utilization Analysis

This section reports the computational cost of running NavA11y, because a tool that is too slow to run on real pages would not be adopted in practice.

The dominant cost is browser-side work, not the analysis logic. For each tabbable element, NavA11y captures a CSS snapshot before and after focus and probes a 7×7 grid of points to measure obscuration. The shape-aware refinement of the obscuration test adds only constant-time geometric arithmetic per element, so it does not change the order of growth; the cost is dominated by the browser’s own layout and paint pipeline, which is unchanged. Across D_r , the tool performed 9,544 element checks over 25 sites, an average of about 382 element checks per site, with the busiest site (walmart.com) requiring 880 checks. The number of element checks is a faithful proxy for the work done, because each check corresponds to one focus event and one measurement pass.

1.8 Discussion of Findings

This section interprets the results rather than merely restating them. We explain why the tool behaved as it did, what the real-world findings mean, and how they connect back to the research questions.

The perfect score on D_d is best read as a validation of correctness on the cases the tool is designed for, not as a claim of universal perfection. The labelled dataset was built to exercise each criterion with clear, unambiguous violation and conforming cases, and the result shows that the detection logic is sound and free of obvious false positives or misses on those cases. The honest counterpart to this strength is stated in Section 1.6 and Section 1.9: the sample is

small, so the confidence intervals are wide, and the perfect score should be reported with that caveat.

The real-world findings carry the more striking message. Focus-behavior violations are pervasive: every one of the 25 major websites tested had at least one violation, and most had dozens. The strong dominance of SC 2.4.13 (Focus Appearance) is explained by design choices that are extremely common on the modern web. Many sites suppress the browser’s default focus outline for aesthetic reasons and either provide no replacement or provide a replacement that is too thin or too low in contrast to meet the AAA appearance bar. Because this is a styling decision applied site-wide, it tends to fail on many elements at once, which is exactly why SC 2.4.13 produces such large counts.

The high counts for the two obscuration criteria (SC 2.4.11 and SC 2.4.12) are explained by the prevalence of sticky headers, fixed footers, cookie banners, and floating chat widgets on commercial sites. These overlays cover focused elements as the user tabs through the page, and SC 2.4.12 in particular fails on any obscuration at all, so even small overlaps are counted. This finding directly motivates the shape-aware obscuration measurement: measuring obscuration against the true rendered shape of an element, rather than its rectangular bounding box, prevents the tool from raising spurious obscuration failures on rounded or non-rectangular buttons, which keeps the large SC 2.4.12 counts trustworthy.

Taken together, the two experiments answer the research questions. RQ-1 is answered by the labelled results: NavA11y detects known focus-behavior violations correctly, with no false positives on the tested cases. RQ-2 is answered by the real-world results and the comparative analysis: the violations are common, they are largely invisible to existing static tools, and NavA11y surfaces them at scale. The practical implication is that automated focus-behavior testing is both necessary, because the problems are everywhere, and currently unserved by mainstream tools, which is the gap this work fills.

1.9 Threats to Validity and Limitations

No study is without limitations, and naming them honestly strengthens rather than weakens the work. We group the threats by type.

A first limitation concerns the labelled dataset. D_d contains 22 synthetic pages, which is a small sample, and the pages were authored partly by us. This creates two risks: the confidence intervals on the perfect detection rate are wide (Section 1.6), and there is a possibility that the contributed pages unconsciously favour the tool’s logic. We mitigated this by building the contributed pages from the established WCAG techniques and failure references rather than from the tool’s implementation, but the risk cannot be fully eliminated. A related, purely practical point concerns reproducibility of the original GDS pages: they were authored to load a shared external stylesheet, and one loads an external script, and these asset files must be present for the intended violation to render. The CSS that removes the focus outline lives in that stylesheet, so a page served without it would render as conforming and the violation would not appear. For this reason the dataset is committed together with its asset files; the result reported in Section 1.4.1 is the one measured with the assets present, which is the configuration the pages were authored for.

A second limitation concerns the real-world dataset and the True-Positive Rate. D_r has no ground-truth labels, so the trustworthiness of its 2,780 violations rests on a manual verification of a stratified sample of flagged cases, reported as a True-Positive Rate. The reliability of that figure depends on the size and representativeness of the sample and on the judgement of the human reviewer.

A third limitation concerns coverage of the page. NavA11y tests a single rendered page per site (typically the home page) and only the elements reachable by the Tab key after the page has hydrated. It does not click, hover, or open menus, lightboxes, and modals, so violations

that only appear inside dynamically revealed content are not exercised. The tool also cannot flag elements that *should* be focusable but are not, because such elements never enter the tab sequence; these are genuine accessibility problems that fall outside the five criteria NavA11y implements.

A fourth limitation concerns the corpus itself. Three of the original top-30 sites were excluded because of automated access barriers, and a small number of others failed at runtime during benchmarking; the reported corpus is therefore the subset that could be evaluated automatically, which may bias the sample slightly toward more automation-friendly sites. Finally, SC 2.4.3 has very few positive cases in D_r , so its real-world prevalence figure should be read as indicative rather than conclusive.

1.10 Summary

This chapter evaluated NavA11y on two complementary datasets and against four established tools. On the labelled dataset, NavA11y achieved 100% precision and 100% recall across all five focus-behavior criteria (32 verdicts), with no false positives and no false negatives, demonstrating that its detection logic is correct on known cases. On 25 real-world production websites, it performed 9,544 element checks and detected 2,780 focus-behavior violations, finding at least one violation on every site, with SC 2.4.13 (Focus Appearance) accounting for more than half of all violations. The comparative analysis showed that the three runtime criteria NavA11y evaluates are not covered at all by mainstream static tools, and a McNemar test confirmed this coverage advantage is statistically significant. We were careful to report these strengths together with their limitations: the labelled sample is small, the real-world trust figure rests on manual sampling, and the tool tests one rendered page per site. The next chapter builds on these results to draw the overall conclusions of the thesis and to outline directions for future work.